
Pair Spacing Reshapes Finite-Budget Learning in Easy Consistency Tuning

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Paper under double-blind review

Abstract

Easy Consistency Tuning (ECT) trains one-step generators from states paired at a temporal separation, but the optimization meaning of that pair spacing is unclear. We show that changing the spacing is an exact composite intervention: it moves the detached target endpoint and changes the explicit inverse-gap weight. These two channels factorize at a matched objective-gradient state, yet the factorization does not determine outcomes after training arms acquire different parameters and optimizer histories. We test this distinction in a fixed- $q = 256$ CIFAR-10 ECT configuration using a controlled $g = 1.10$ spacing probe. A three-seed formal factorial favored the complete intervention over the baseline in all 12 seedwise FID/KID directions across one- and two-step sampling at 256 king. Audited exact-budget replays and two secondary cohorts then track 456 paired FID/KID evaluation jobs through 1024 king. At the descriptive one-step FID threshold of 10, the complete intervention reached the threshold one 128-king stage earlier for five of ten training seeds; two seeds tied, two crossed later, and one pair was censored. Four-arm rankings also changed with seed, budget, and sampling NFE. Pair spacing can therefore advance entry into a useful finite-budget regime, but the advance is not universal. In this setting, pair spacing is best understood as a locally structured training control with trajectory- and horizon-dependent consequences, not as a universally superior final hyperparameter.

1 Introduction

Consistency models replace iterative diffusion sampling with a mapping that is constant along a probability-flow trajectory, enabling one- or few-step generation (Song et al., 2023). Their sampling speed shifts part of the computational burden to training: a useful model must learn a long trajectory relation from local or discretized consistency constraints. Improved consistency training and Easy Consistency Tuning (ECT) make this process far more effective (Song & Dhariwal, 2024; Geng et al., 2024), but they also introduce a concrete control question. How far apart should the two training states be? This question matters at finite budget because pair spacing can change when a one-step model first enters a useful-quality regime, even if differences later contract.

Pair spacing is easy to misread as a scalar knob. In the ECT objective studied here, a larger spacing changes which earlier state supplies the detached target and also changes the explicit inverse-gap weight multiplying the residual. These are distinct interventions. At one matched model state, their action on the loss and one-sided training derivative is exactly characterizable. After a single update, however, separately trained arms generally no longer share parameters, stochastic gradients, or optimizer state. A local identity is therefore not an endpoint-ranking theorem. This local-to-trajectory distinction is the organizing problem of the paper (Figure 1).

We study this problem at fixed $q = 256$. Both q and the multiplier g can affect realized pair spacing; our experiment changes only g and does not establish invariance across q . The novelty is consequently not that $g = 1.10$ is “better.” It is the conjunction of (i) an exact ECT-specific composite-intervention formulation, (ii) a target-by-denominator factorial fol-

Pair spacing is a composite local intervention, not a global training equivalence

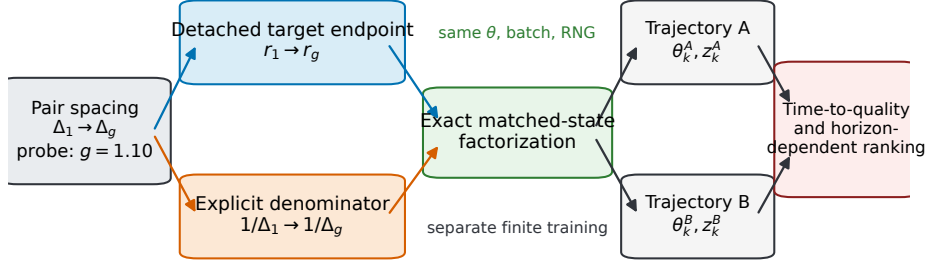


Figure 1: Pair spacing is a composite local intervention. In the studied ECT objective, changing Δ moves the detached target endpoint and changes the explicit inverse-gap weight. Their matched-state loss/gradient action factorizes exactly. Separate finite training then creates different parameter and optimizer histories, so time-to-quality and arm ranking remain empirical, horizon-dependent quantities. The fixed $g = 1.10$ multiplier is a controlled probe, not a proposed universal method.

lowed through complete learning trajectories, and (iii) a seed-resolved account of when the intervention advances, ties, or delays entry into a high-quality one-step regime.

Our contributions are threefold:

1. We derive exact finite-gap loss and one-sided gradient decompositions, exact target-by-denominator identities, and a conditional local scaling law $\kappa = \nu(p-1) - \alpha$ that specializes to $\kappa = -1$ for the ECT loss.
2. We combine a formal three-seed factorial with audited deterministic four-arm replays over seven budgets. This design separates objective-level factorization from finite-training outcome non-factorization.
3. We quantify finite-budget phase advance without averaging away adverse seeds. Across three evidence groups, the complete intervention is one observed stage earlier for five seeds, tied for two, later for two, and censored for one; component rankings also depend on horizon and sampling NFE.

2 Related Work and Problem Setup

2.1 Training fast generative models

Consistency models can be trained by distillation or directly from data (Song et al., 2023). Improved consistency training (iCT) corrected the teacher update and introduced loss, noise-sampling, and discretization choices that substantially strengthened training from scratch (Song & Dhariwal, 2024). ECT instead fine-tunes a pretrained diffusion model while progressively strengthening consistency constraints (Geng et al., 2024). Related work improves continuous-time parameterization and stability (Lu & Song, 2025), reduces target variance (Wang et al., 2024; Silvestri et al., 2025), or adapts traversal difficulty and discretization (Liu et al., 2025; Bai et al., 2025). These methods establish that pair construction, weighting, and trajectory coverage affect trainability. Our question is narrower: for one fixed ECT configuration, what intervention does pair spacing implement locally, and how does that intervention appear along finite training trajectories?

Adaptive optimizers provide a relevant boundary. Adam and RAdam retain unequal moment histories (Kingma & Ba, 2015; Liu et al., 2020), and nonstationary gradient magnitude can interact with stored moments (Ellis et al., 2024). Gradient scaling therefore need not imply trajectory equivalence. Section 6 uses a frozen-state ECT diagnostic only to establish this boundary.

2.2 ECT objective and factorial arms

For a coupled sample, write $x_t = y + t\epsilon$ and $x_r = y + r\epsilon$, with $\Delta = t - r > 0$. The online output is $D_t = f_\theta(x_t, t)$ and the target is $T_r = \text{sg}[f_\theta(x_r, r)]$. The studied loss uses the Euclidean residual and an external inverse-gap weight,

$$\ell(r, \Delta) = \frac{\|D_t - T_r\|_2}{\Delta}. \quad (1)$$

We compare the native baseline spacing Δ_1 with the realized spacing Δ_g under $g = 1.10$, including the production boundary clamp. The four training arms cross the detached target and denominator assignments:

Table 1: Target-by-denominator factorial.

Arm	Target	Denominator	Role
A	$r_1 = t - \Delta_1$	Δ_1	baseline
B	$r_g = t - \Delta_g$	Δ_g	complete probe
C	r_g	Δ_1	target only
D	r_1	Δ_g	denominator only

3 Exact Local Structure of the Intervention

3.1 Ideal endpoint invariance is conditional

Pair spacing can change training without changing an ideal consistency map. Let $\mathcal{T}_{t \rightarrow r}$ be exact compositional transport along a trajectory. Suppose a schedule connects every supported t to a common anchor t_* , every scheduled edge is covered by the population objective, the function class attains zero edge residual, optimization reaches that solution, and the map is fixed at the anchor. Zero residual then telescopes:

$$F^*(x_t, t) = F^*(\mathcal{T}_{t \rightarrow t_*}(x_t), t_*). \quad (2)$$

Two legal schedules with the same support, transport, and anchor therefore identify the same ideal map on reachable states. This is an identification statement under strong assumptions, not a claim about finite-capacity or incompletely optimized networks (Lyu et al., 2024; Dou et al., 2024).

3.2 One-sided derivative and exact finite-gap decomposition

Stop-gradient must be represented at the derivative level. Introduce a lifted loss $\tilde{\ell}(\theta, \bar{\theta})$ whose target branch uses $\bar{\theta}$. The training derivative is

$$\mathcal{D}_\theta^{\text{sg}} \ell := \partial_\theta \tilde{\ell}(\theta, \bar{\theta}) \Big|_{\bar{\theta}=\theta}, \quad (3)$$

not the diagonal total derivative, which would also differentiate the target copy. Let $e_r = D_t - T_r$, $u_r = e_r / \|e_r\|_2$, and $J_t = \partial_\theta D_t$. Away from $e_r = 0$, the implemented per-sample training derivative is exactly

$$g(r, \Delta) := \mathcal{D}_\theta^{\text{sg}} \ell(r, \Delta) = \frac{1}{\Delta} J_t^\top u_r. \quad (4)$$

Evaluate the baseline and enlarged spacing at the same parameters, online realization, data, and noise. With $s = \Delta_1 / \Delta_g$, substitution into Equations (1)–(4) yields

$$\ell_g = s\ell_1 + \frac{\|e_{r_g}\|_2 - \|e_{r_1}\|_2}{\Delta_g}, \quad (5)$$

$$g_g = sg_1 + \frac{1}{\Delta_g} J_t^\top (u_{r_g} - u_{r_1}). \quad (6)$$

The first term is exact denominator rescaling; the second contains the entire target-endpoint change. No small-gap or optimizer approximation is used.

The factorial follows immediately. At a matched sample,

$$g_B = sg_C, \quad g_D = sg_A, \quad g_B - g_D = s(g_C - g_A), \quad (7)$$

and the interaction satisfies

$$g_B - g_C - g_D + g_A = (s - 1)(g_C - g_A). \quad (8)$$

When the multiplier is unclipped, $s = 1/g$ is common and these identities also hold after minibatch summation. The production manipulation test verified the two scalar identities with maximum elementwise relative residuals below 3.9×10^{-7} . These are local objective-gradient identities; they neither rank trained arms nor decompose FID into causal percentages.

3.3 A conditional local scaling law

The exact decomposition has a useful asymptotic corollary. Consider $\ell_\delta = w(\delta, z)\Phi(e_\delta)$ and assume $e_\delta = \delta^\nu v + O(\delta^{\nu+1})$ with $v \neq 0$, $w(\delta, z) = w_0(z)\delta^{-\alpha}(1 + O(\delta))$, a regular online Jacobian, a nonzero leading derivative, and a degree- p homogeneous Φ . Then

$$g_\delta^{\text{sg}} = C(\theta, z)\delta^\kappa + O(\delta^{\kappa+1}), \quad \boxed{\kappa = \nu(p - 1) - \alpha}. \quad (9)$$

For the present $c = 0$ ECT loss, $p = 1$ and $\alpha = 1$, hence $\kappa_{\text{ECT}} = -1$. This special case predicts a leading inverse-gap scale, independent of ν ; it therefore does not validate the residual order ν . It is also local. The finite target correction in Equation (6), minibatch heterogeneity, and all subsequent optimizer states remain outside the scalar prediction. Appendix A gives the proof and scope.

4 Experimental Design and Evidence Status

Training and evaluation. We fine-tuned the same unconditional CIFAR-10 32×32 DDPM++ model with ECT preconditioning in every arm. Runs used $q = 256$, RAdam with learning rate 10^{-4} and $(\beta_1, \beta_2) = (0.9, 0.999)$, global batch 128, dropout 0.2, EMA decay 0.9993, FP16 networks with AMP, and no augmentation or weight decay. The formal arms began from the same immutable pretrained checkpoint within each seed and differed only in the target/denominator assignment in Table 1. FID and KID use 50,000 generated samples, generation seeds 0–49,999, metric seed 20260730, and FP32 sampling (Heusel et al., 2017; Bińkowski et al., 2018). NFE2 uses the frozen intermediate time 0.821. Training seed is the independent unit; budgets, NFE, and the two metrics are repeated readouts.

Evidence groups. The original formal experiment contains seeds 3–5, four arms, and the 256-king endpoint. A deterministic replay of the same 12 trajectories provides exact checkpoints at 256, 384, 512, 640, 768, 896, and 1024 king; all 168 FID/KID jobs passed, and all 12 terminal replay states were bitwise identical to their canonical 1024-king states. Secondary evidence is reported separately: seeds 6–7 contain A/B curves from 384 to 1024 king (48 jobs), and seeds 14–18 contain four-arm curves over the same six budgets (240 jobs). These secondary cohorts improve descriptive coverage but are not prospectively relabelled confirmation.

Time-to-quality. For arm a , seed s , and descriptive threshold $\eta = 10$, define

$$\tau_{a,s}(\eta) = \min\{K : \text{FID}_{a,s}^{\text{NFE1}}(K) \leq \eta\}. \quad (10)$$

We use observed checkpoints only and report censoring when a crossing is absent. The threshold summarizes entry into the quality range reached late in training; it was not pre-registered as a confirmatory endpoint. Matched-budget differences are reported alongside it so the conclusion does not rest on one threshold.

5 Finite-Budget Learning Results

5.1 The controlled early endpoint favors the complete intervention

At 256 king, B had lower FID and KID than A for every formal seed under both NFE1 and NFE2 (Table 2). This gives 12 favorable seedwise directions: three seeds, two metrics, and two samplers. It is an early finite-budget intervention result, not a practical terminal benchmark; among A and B, NFE1 FID remained between 306.8 and 331.9.

Table 2: Formal 256-king seed-level results. FID columns show all four NFE1 arms. Δ is B–A; negative values favor the complete intervention. KID differences are multiplied by 10^3 . FID and KID share generated features and are not independent experiments.

Seed	NFE1 FID-50k				NFE1 B–A		NFE2 B–A	
	A	B	C	D	Δ FID	$10^3\Delta$ KID	Δ FID	$10^3\Delta$ KID
3	331.885	310.508	318.175	329.234	-21.377	-39.895	-138.277	-164.555
4	310.235	306.773	304.566	324.722	-3.461	-5.470	-0.924	-1.640
5	318.639	308.855	308.414	315.200	-9.784	-3.309	-5.030	-6.032

The factorial qualifies the interpretation. At NFE1, both target-related contrasts, C–A and B–D, were negative for all three seeds. By contrast, the denominator contrasts D–A and B–C and the factorial interaction had mixed signs. This pattern motivated the component analysis, but it does not show that the target explains a fixed share of B–A or that the denominator is irrelevant.

5.2 Exact-budget curves reveal contraction and rank changes

The deterministic replay replaces a single endpoint with the complete formal learning paths (Figure 2). Early separations can be large: for example, the NFE1 B–A FID difference at 384 king is -65.65 for seed 5. The sign reverses at 512 king ($+5.61$) and becomes favorable again after 640 king. Seed 4 changes in the other direction: B is favorable through 512 king and slightly unfavorable thereafter. Thus even within one seed, matched-budget direction is not fixed by the early endpoint.

By 1024 king, the formal NFE1 B–A differences are -0.983 , $+0.229$, and -0.596 for seeds 3–5. The corresponding NFE2 differences are -0.034 , $+0.023$, and -0.101 . Absolute quality is now substantially better—formal NFE1 FID is 7.64–10.67 across all arms and NFE2 FID is 2.71–3.09—but the best arms are B, A, and C for both samplers across the three seeds. Mature contraction therefore does not produce a universal terminal winner.

5.3 Phase advance occurs on several, not all, trajectories

Figure 3 reports the first observed NFE1 checkpoint with FID at most 10. B crosses one 128-king stage earlier for seeds 3, 5, 6, 7, and 14. Seeds 4 and 15 tie; seeds 17 and 18 cross one stage later under B. Neither arm crosses for seed 16 by 1024 king (A: 13.13; B: 10.92), so that pair is censored. The phase advance is consequently real on multiple trajectories but not a seed-uniform property.

Matched-budget comparisons tell the same, less threshold-dependent story. B has lower NFE1 FID at 16/21 formal replay points, 9/12 secondary A/B points, and 20/30 secondary factorial points. These are repeated observations along ten trajectories, not 45 independent wins. At 1024 king the favorable B–A counts are 2/3, 2/2, and 3/5 in the three evidence groups, respectively. The effect therefore persists into a mature-quality regime for several seeds, but its direction is neither universal nor constant over horizon.

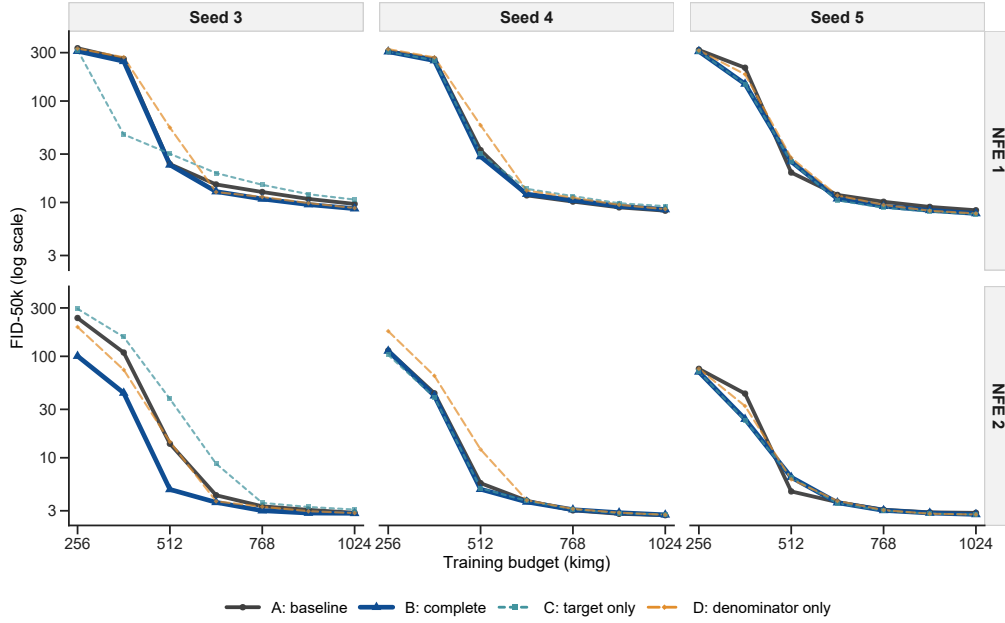


Figure 2: Formal deterministic four-arm learning curves. Every panel shows all arms for one training seed at the seven exact replay budgets. FID is plotted on a log scale; NFE and budget are repeated readouts of the same three seed trajectories. Large early differences contract, while the order of A/B/C/D remains seed- and NFE-dependent.

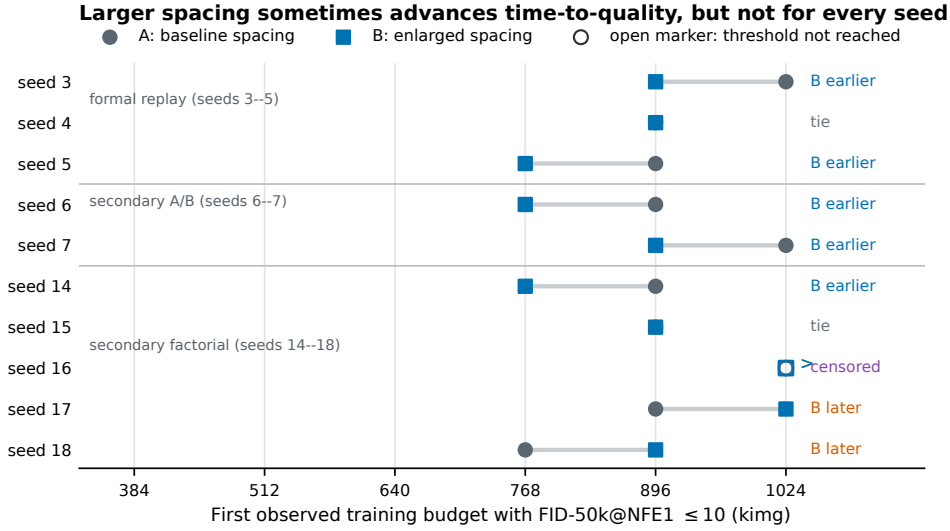


Figure 3: Seed-resolved descriptive time-to-quality. Points are the first observed budgets with $\text{FID-50k@NFE1} \leq 10$; connectors pair A and B within a training seed. Evidence groups are separated. Open markers denote no crossing by the last observed checkpoint. The grid is 128 king and no interpolation is used. The threshold was chosen for descriptive analysis, not as a preregistered confirmatory endpoint.

5.4 Component rankings are horizon- and sampler-dependent

The formal contrast paths make outcome non-factorization explicit (Figure 4). At 256 king, both NFE1 target contrasts are negative for all three seeds. At 1024 king, C–A is positive

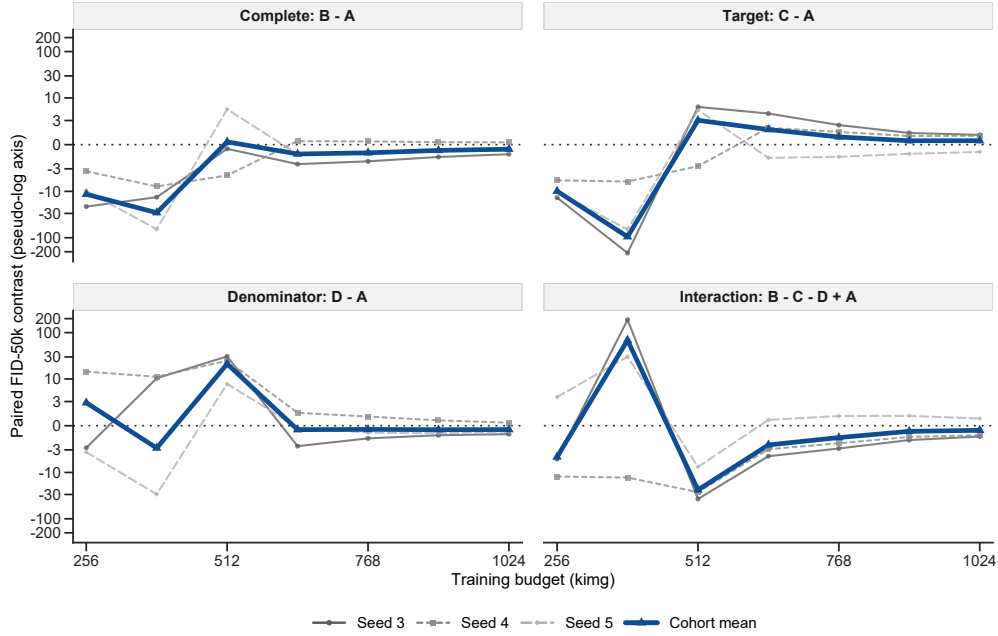


Figure 4: Formal NFE1 FID contrasts across budget. Negative values favor the first arm in each title. A signed pseudo-log axis preserves zero and displays both early large effects and late small contrasts. Thin lines are training seeds; the thick line is their descriptive mean. Target, denominator, and interaction directions change with seed and horizon.

for seeds 3 and 4 but negative for seed 5; denominator and interaction contrasts also mix signs. The exact relation in Equation (8) still holds for a matched objective state, but it does not constrain these separately trained FID contrasts.

The five-seed secondary factorial supplies an adverse-sensitivity check. At 1024 kimg, the NFE1 winners for seeds 14–18 are D, B, B, A, and A; the NFE2 winners are D, A, D, A, and A. Thus the sampler changes the winner for two of five seeds, while no single arm wins across the cohort. Early rankings differ again. The empirical conclusion is therefore narrower than a component hierarchy: the intervention factorizes exactly at the objective level, while the functional ranking of its components depends on budget, seed, and sampler.

6 Supporting Optimizer-State Boundary

Stateful optimization is one reason a local scalar relation need not become a trajectory equivalence. We compared virtual RAdam updates after best scalar alignment,

$$R_{\text{opt}}(U_1, U_0) = \frac{\|U_1 - c^* U_0\|_2}{\|U_0\|_2}, \quad c^* = \frac{\langle U_1, U_0 \rangle}{\|U_0\|_2^2}. \quad (11)$$

Across seeds 3–5, the median residual was 0.074–0.088 for observed gradients under real accumulated state and 0.467–0.491 after moment reset. Even an exact-scalar gradient pair retained 0.049–0.077 under real state. Moment reset is therefore not a memory-neutral control. A preregistered moment-transport attempt subsequently worsened all 12 held-out residuals and failed its gate; no continuation training or FID/KID evaluation was launched. These diagnostics support only the boundary that instantaneous gradients do not determine stateful updates. They do not establish that RAdam caused the quality changes.

7 Discussion, Limitations, and Conclusion

Scientific interpretation. Pair spacing in ECT is locally structured but globally trajectory-dependent. The exact decomposition explains why the intervention cannot be reduced to a single unspecified “gap effect.” The curves explain why the decomposition should not be promoted to a fixed quality attribution: each arm changes the future states at which later objectives are evaluated. The useful practical quantity is therefore not only terminal FID, but the path from compute to quality. On five trajectories, the complete probe advances the first observed high-quality NFE1 checkpoint by one stage. Ties, delays, censoring, and changing four-arm winners establish the boundary of that advance.

Limitations and next tests. The evidence comes from CIFAR-10, one unconditional DDPM++/ECT configuration, fixed $q = 256$, and one spacing multiplier. Both g and q affect effective pair spacing, so transfer across q is unknown. NFE1 and NFE2 can rank arms differently, the time-to-quality threshold is descriptive, and the extended cohorts are secondary/post-preregistration analyses. The optimizer diagnostic does not close a causal path to FID, and ideal endpoint invariance depends on the assumptions preceding Equation (2). The highest priority validation is an effective-spacing-matched cross- q control, followed by a fresh prospective phase-advance test and a second weighting rule or dataset. Those tests would determine whether trajectory-conditioned scheduling is warranted; simply adding more retrospective $q = 256$ seeds would not resolve the main generalization question.

Conclusion. Changing ECT pair spacing exactly combines a detached-target intervention with inverse-gap reweighting. This local factorization coexists with finite-budget learning paths whose quality and component rankings depend on seed, horizon, and sampler. Pair spacing can advance time-to-quality, but it does not define a universally best static gap. The resulting view is a bounded one: pair spacing is a finite-budget training control whose local action can be derived and whose global consequence must be measured along the trajectory.

AI Use Statement

Generative AI tools assisted hypothesis refinement, experiment-design feedback, code generation and editing, code and derivation review, theoretical auditing, result interpretation, manuscript organization, prose editing, and figure-code preparation. Numerical measurements reported in the paper were produced by executed evaluation pipelines, not generated by a language model. The authors checked formulas, numerical transcriptions, figures, and claims against the archived repository records, reviewed all AI-assisted content, and assume responsibility for the manuscript.

Ethics Statement

This work studies optimization on the public CIFAR-10 benchmark and introduces no human-participant study, private data, or deployment system. Its main risk is misinterpretation of a configuration-specific diagnostic as a universal training prescription; the evidence-status labels and limitations are intended to prevent that use.

Reproducibility Statement

The supplementary package includes exact arm definitions, source-bound seed-level metric tables, deterministic replay checks, figure-generation code, and compact CSV/JSON artifacts. Full training and evaluation settings are listed in Appendix D. Checkpoints and generated samples are excluded from the paper package because of size, but their hashes are retained in the audit records.

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A Proof and Scope of the Local Scaling Law

For the lifted one-sided objective, the exact chain rule is $g_\delta^{\text{sg}} = w(\delta, z)J_\delta^{+\top} \nabla \Phi(e_\delta)$. Under the assumptions in Section 3, positive homogeneity gives

$$\nabla \Phi(e_\delta) = \delta^{\nu(p-1)} [\nabla \Phi(v) + O(\delta)]. \quad (12)$$

Multiplying by $w_0 \delta^{-\alpha} [1 + O(\delta)]$ and $J_\delta^{+\top} = J_0^{+\top} + O(\delta)$ proves Equation (9), provided $J_0^{+\top} \nabla \Phi(v) \neq 0$. The theorem does not cover the ordinary diagonal total derivative, large gaps, a Pseudo-Huber crossover, a non-smooth clipping boundary, or a degenerate leading term. The exact finite-gap identities in Equations (6)–(8) do not require this asymptotic theorem.

B Full Formal Endpoint Matrix

Table 3: Formal 256-king metrics. Lower is better.

Seed	Arm	FID ₁	KID ₁	FID ₂	KID ₂
3	A	331.885	.353531	238.774	.252762
3	B	310.508	.313637	100.498	.088207
3	C	318.175	.343722	294.103	.312204
3	D	329.234	.345016	194.670	.202756
4	A	310.235	.315593	113.817	.103911
4	B	306.773	.310123	112.893	.102271
4	C	304.566	.307218	103.090	.091747
4	D	324.722	.339143	176.961	.180122
5	A	318.639	.329435	75.266	.062150
5	B	308.855	.326125	70.236	.056118
5	C	308.414	.325736	70.486	.056508
5	D	315.200	.329756	75.408	.062109

C Secondary Trajectory Summaries

Table 4: First observed budget with NFE1 FID ≤ 10 . A dash denotes no crossing by 1024 king.

Seed	Evidence group	τ_A	τ_B	Class
3	formal replay	1024	896	B earlier
4	formal replay	896	896	tie
5	formal replay	896	768	B earlier
6	secondary A/B	896	768	B earlier
7	secondary A/B	1024	896	B earlier
14	secondary factorial	896	768	B earlier
15	secondary factorial	896	896	tie
16	secondary factorial	—	—	censored
17	secondary factorial	896	1024	B later
18	secondary factorial	768	896	B later

For the secondary factorial at 1024 king, NFE1 winners for seeds 14–18 are D/B/B/A/A, while NFE2 winners are D/A/D/A/A. The corresponding NFE1 B–A differences are -1.266 , -0.072 , -2.213 , $+1.248$, and $+0.934$. These values are descriptive sensitivity evidence and are not pooled with the formal seed cohort.

D Protocol and Optimizer Diagnostic Details

The training loss used $q = 256$, $k = 8$, $b = 1$, $c = 0$, a global batch of 128 with per-device minibatches of 16, RAdam learning rate 10^{-4} , betas (0.9, 0.999), epsilon 10^{-8} , zero weight decay, dropout 0.2, no data augmentation or horizontal flip, FP16 networks, AMP loss scaling 1.0, TF32 disabled, and EMA decay 0.9993. Formal training used deterministic algorithms, deterministic cuDNN, cuDNN benchmarking disabled, and CUBLAS_WORKSPACE_CONFIG=:4096:8. The archived execution environment records Python 3.10.12, PyTorch 2.2.0a0+81ea7a4, CUDA 12.3, cuDNN 8.9.0.7, and two NVIDIA A100 80GB PCIe devices for the training queues.

The frozen-state optimizer factorial uses three training seeds and eight paired audit minibatches per seed. The latter are repeated measurements. Its four median- R_{opt} cells (observed/real, observed/reset, exact-scalar/real, exact-scalar/reset) are respectively (0.0830, 0.4770, 0.0704, 0.00171), (0.0878, 0.4907, 0.0772, 0.00164), and (0.0741, 0.4669, 0.0489, 0.00060) for seeds 3–5. The subsequent held-out moment-transport gate produced negative suppression for every seed and all 12 minibatches, so the planned continuation stopped before training.